FR-03 – Step 3: Select test cases (best representative of each class)

Input: the classes in `FR-03_2_equivalence_classes.md`. Method: lecture S04, step 3: - Choose **at least one** test case from each equivalence class. - **Valid** classes: each test case covers as many valid classes as possible. - **Invalid** classes: each test case covers **one and only one** invalid class. All its other inputs come from valid classes, so a failure points to exactly one cause.

Columns follow the course template `Test cases_Template_En.xlsx`. Actual result, status, tester and date are filled in at step 5 (execution).

Common setup

ID	Setup
P1	Backend (`:3000`) and frontend-web (`:5173`) running. Database freshly seeded with `node database.js` , so `test@eshop.com` / `Test1234!` exists.
P2	A second account, **user B**, registered through the Register page: `userb@eshop.com` / `UserB123!` .
P3	**Step 2 ready:** open `http://localhost:5173/forgot-password` , enter `test@eshop.com` , submit. The page shows step 2 and an OTP; call it **X**.
API	API test cases use Postman or curl. `POST http://localhost:3000/api/forgot-password` returns the OTP in `resetToken` (API spec 1.3). `POST http://localhost:3000/api/reset-password` takes `email` , `resetToken` , `newPassword` (API spec 1.4).
Cleanup	After any test case that changes the password (expected or not), re-seed the database and redo P2. This keeps every test case independent of the others.

Default valid values used whenever a test case doesn't target that field: email `test@eshop.com`, OTP = **X**, new password `NewPass1!` , confirm `NewPass1!` .

Valid test cases

TC ID	Test case name	Classes covered	Precondition	Test steps (with data)	Expected result
FR-03_TC_01	Reset password with all-valid inputs (happy path)	EC_01, 03, 07, 13, 15, 18, 21, 24, 26, 29, 31, 33, 35, 40, 45, 47, 50, 54, 60; outputs EC_62, 63, 64, 74, 75, 77	P1	1. Open forgot-password page; check the step indicator. 2. Email `test@eshop.com` , submit. 3. Note OTP X; check the step indicator. 4. OTP `X` , new password `NewPass1!` , confirm `NewPass1!` , submit within 1 minute. 5. Log in with `NewPass1!` . 6. Log out, log in with `Test1234!` .	1. "Bước 1 / 2" shown. 3. 6-digit OTP shown; "Bước 2 / 2" shown. 4. Success message; redirected to login. 5. Login succeeds. 6. Login fails.

TC ID	Test case name	Classes covered	Precondition	Test steps (with data)	Expected result
FR-03_TC_02	"Back to login" on step 1	EC_11; output EC_76	P1	1. Open forgot-password page. 2. Click "Back to login".	Login page is shown.
FR-03_TC_03	"Back to login" on step 2	EC_12; output EC_76	P3	1. On step 2, click "Back to login".	Login page is shown.
FR-03_TC_04	Use the latest of two OTPs; new password equals current	EC_55, EC_39	P1	1. Request an OTP for `test@eshop.com` (OTP X1). 2. Request again (OTP X2). 3. OTP `X2`, new password `Test1234!`, confirm `Test1234!`, submit.	Success message; login with `Test1234!` works.
FR-03_TC_05	OTP typed with surrounding spaces	EC_20	P3	1. OTP `" X "` (one space before and after), defaults for the rest, submit.	Success (spaces are trimmed).
FR-03_TC_06	Correct OTP that starts with 0	EC_23	P1	1. Repeat step 1 until the OTP shown starts with `0` (about 1 try in 10). 2. Enter that OTP exactly, defaults for the rest, submit.	Success.
FR-03_TC_07	Reset API with a matching confirm field	EC_57	P1	1. Get OTP X via API. 2. `POST /api/reset-password {"email":"test@eshop.com","resetToken":"X","newPassword":"NewPass1!","confirmPassword":"NewPass1!"}`	HTTP 200; password changed.

Invalid test cases – step 1 (email)

Precondition for all: P1. Each submits only the email shown.

TC ID	Test case name	Class covered	Test steps (with data)	Expected result
FR-03_TC_08	Empty email	EC_02; output EC_67	Email `""` (also try `" "`), submit.	Required-field error; stays on step 1; no OTP.
FR-03_TC_09	Email without `@`	EC_04; output EC_66	Email `testeshop.com`, submit.	Invalid-format error; no OTP.
FR-03_TC_10	Email with empty local part	EC_05	Email `@eshop.com`, submit.	Invalid-format error; no OTP.
FR-03_TC_11	Email without top-level domain	EC_06	Email `test@eshop`, submit.	Invalid-format error; no OTP.
FR-03_TC_12	Well-formed but unregistered email	EC_08; output EC_65	Email `nobody@eshop.com`, submit.	"Not registered" error; no OTP.

TC ID	Test case name	Class covered	Test steps (with data)	Expected result
FR-03_TC_13	Registered email in different case	EC_09	Email `Test@eshop.com`, submit.	"Not registered" error (tester decision).
FR-03_TC_14	Registered email with surrounding spaces	EC_10	Email `" test@eshop.com "`, submit.	Invalid-format error (tester decision).

Invalid test cases – step 2 (OTP)

Precondition for all: P3. New password and confirm are `NewPass1!` unless stated.

TC ID	Test case name	Class covered	Test steps (with data)	Expected result
FR-03_TC_15	Empty OTP	EC_14	OTP `""`, submit.	Required-field error.
FR-03_TC_16	OTP shorter than 6	EC_16; output EC_68	OTP = first 5 digits of X, submit.	Incorrect-OTP error.
FR-03_TC_17	OTP longer than 6	EC_17	OTP = X followed by `1` (7 digits), submit.	Incorrect-OTP error.
FR-03_TC_18	OTP containing a letter	EC_19	OTP = X with its 3rd digit replaced by `a`, submit.	Incorrect-OTP error.
FR-03_TC_19	Wrong 6-digit OTP; password unchanged	EC_22; output EC_78	1. OTP = X with its last digit changed (e.g. +1, 9→0), submit. 2. Log in with `Test1234!`.	1. Incorrect-OTP error. 2. Login succeeds (password unchanged).
FR-03_TC_20	Expired OTP	EC_46; output EC_69	Wait **15 minutes** after getting X, then submit with OTP X. (T is not in the spec; 15 min is assumed to be past any reasonable expiry.)	OTP-expired error.
FR-03_TC_21	Reuse an OTP after a successful reset	EC_48; output EC_70	1. Submit OTP X with `NewPass1!` (succeeds). 2. Request nothing new; open step 2 again (or use the API) with OTP X, password `Other123!`.	2. OTP-already-used error; `Other123!` does not log in.
FR-03_TC_22	Reuse an OTP after a failed reset	EC_49	1. Submit OTP X with password `short 1!` (fails, too weak). 2. Submit OTP X again with `NewPass1!`.	2. OTP-already-used error (tester decision).
FR-03_TC_23	Older OTP after a newer one was requested	EC_56	1. Request OTP X1, then request OTP X2. 2. Submit OTP `X1`.	Incorrect-OTP error.

Invalid test cases – step 2 (new password)

Precondition for all: P3. OTP = X; confirm = the same value as the new password, so only the password rule is wrong.

TC ID	Test case name	Class covered	New password and confirm	Expected result
FR-03_TC_24	Empty password	EC_25	`""`	Required-field error.

TC ID	Test case name	Class covered	New password and confirm	Expected result
FR-03_TC_25	Too short (7)	EC_27; output EC_72	`Pass12!`	Weak-password error.
FR-03_TC_26	Very long (129)	EC_28	`Aa1!` + 125 × `a`	Weak-password / length error (tester decision); no crash.
FR-03_TC_27	No uppercase	EC_30	`newpass1!`	Weak-password error.
FR-03_TC_28	No lowercase	EC_32	`NEWPASS1!`	Weak-password error.
FR-03_TC_29	No digit	EC_34	`NewPass!!`	Weak-password error.
FR-03_TC_30	No special character	EC_36	`NewPass12`	Weak-password error.
FR-03_TC_31	Special character only outside the set	EC_37	`NewPass1#`	Weak-password error.
FR-03_TC_32	Contains a space	EC_38	`New Pass1!`	Weak-password error (tester decision).

Invalid test cases – step 2 (confirm password)

Precondition for all: P3. OTP = X; new password = `NewPass1!`.

TC ID	Test case name	Class covered	Confirm value	Expected result
FR-03_TC_33	Empty confirm	EC_41	`""`	Required-field error.
FR-03_TC_34	Completely different	EC_42; output EC_73	`Other123!`	Passwords-don't-match error.
FR-03_TC_35	Differs only in case	EC_43	`newpass1!`	Passwords-don't-match error.
FR-03_TC_36	Differs only by a trailing space	EC_44	`"NewPass1! "`	Passwords-don't-match error.

Invalid test cases – reset API

Precondition for all: P1 + P2. Get OTP X for `test@eshop.com` via `POST /api/forgot-password` unless stated. Each request uses valid values except the one field named.

TC ID	Test case name	Class covered	Request body for `POST /api/reset-password`	Expected result
FR-03_TC_37	OTP of user A used for user B	EC_51; output EC_71	Also request an OTP for user B. Then `{"email":"userb@eshop.com","resetToken":"X","newPassword":"NewPass1!"}`	Error "OTP not valid for this email"; user B's password unchanged.
FR-03_TC_38	Unregistered email	EC_52	`{"email":"nobody@eshop.com","resetToken":"X","newPassword":"NewPass1!"}`	Error; no account changed.
FR-03_TC_39	Email missing	EC_53	`{"resetToken":"X","newPassword":"NewPass1!"}`	Error; no account changed.

TC ID	Test case name	Class covered	Request body for `POST /api/reset-password`	Expected result
FR-03_TC_40	Confirm field present but not matching	EC_58	`{"email":"test@eshop.com","resetToken":"X","newPassword":"NewPass1!","confirmPassword":"Other123!"}`	Passwords-don't-match error; password unchanged.
FR-03_TC_41	Confirm field absent	EC_59 (TBD)	`{"email":"test@eshop.com","resetToken":"X","newPassword":"NewPass1!"}` (exactly API spec 1.4)	**Decided at execution.** Record what happens; the spec conflict is reported either way.
FR-03_TC_42	Reset for an email that never requested an OTP	EC_61	Do **not** request an OTP for user B. `{"email":"userb@eshop.com","resetToken":"123456","newPassword":"NewPass1!"}`	Error; user B's password unchanged.

Coverage check

Every class from step 2 is covered by at least one test case, and each invalid test case targets exactly one invalid input class.

Class	Test case		Class	Test case		Class	Test case
EC_01	TC_01		EC_27	TC_25		EC_53	TC_39
EC_02	TC_08		EC_28	TC_26		EC_54	TC_01
EC_03	TC_01		EC_29	TC_01		EC_55	TC_04
EC_04	TC_09		EC_30	TC_27		EC_56	TC_23
EC_05	TC_10		EC_31	TC_01		EC_57	TC_07
EC_06	TC_11		EC_32	TC_28		EC_58	TC_40
EC_07	TC_01		EC_33	TC_01		EC_59	TC_41
EC_08	TC_12		EC_34	TC_29		EC_60	TC_01
EC_09	TC_13		EC_35	TC_01		EC_61	TC_42
EC_10	TC_14		EC_36	TC_30		EC_62	TC_01
EC_11	TC_02		EC_37	TC_31		EC_63	TC_01
EC_12	TC_03		EC_38	TC_32		EC_64	TC_01
EC_13	TC_01		EC_39	TC_04		EC_65	TC_12
EC_14	TC_15		EC_40	TC_01		EC_66	TC_09
EC_15	TC_01		EC_41	TC_33		EC_67	TC_08
EC_16	TC_16		EC_42	TC_34		EC_68	TC_16
EC_17	TC_17		EC_43	TC_35		EC_69	TC_20
EC_18	TC_01		EC_44	TC_36		EC_70	TC_21
EC_19	TC_18		EC_45	TC_01		EC_71	TC_37
EC_20	TC_05		EC_46	TC_20		EC_72	TC_25
EC_21	TC_01		EC_47	TC_01		EC_73	TC_34
EC_22	TC_19		EC_48	TC_21		EC_74	TC_01

Class	Test case		Class	Test case		Class	Test case
EC_23	TC_06		EC_49	TC_22		EC_75	TC_01
EC_24	TC_01		EC_50	TC_01		EC_76	TC_02
EC_25	TC_24		EC_51	TC_37		EC_77	TC_01
EC_26	TC_01		EC_52	TC_38		EC_78	TC_19

Summary

	Count
Valid test cases	7
Invalid test cases	34
Test case with expected result decided at execution	1 (TC_41)
Total test cases	**42**
Classes covered	78 / 78

Notes for review

1. **TC_01 carries 19 valid input classes.** That is what the lecture asks for, but if it fails, the cause is unclear. If TC_01 fails at execution, split it before reporting the bug.
2. **TC_04 combines two valid classes** (latest OTP, password equal to current). If it fails, rerun each part on its own.
3. **TC_06 depends on luck:** the OTP must happen to start with `0`. If it hasn't after 30 tries, mark it "Not executed" with that reason.
4. **TC_20 assumes T ≤ 15 minutes.** If the OTP still works after 15 minutes, that is still a finding: SEC-07 says the OTP must expire.
5. **TC_07, TC_40, TC_41 send a `confirmPassword` field** whose name is assumed: the API spec has no such field.
6. **Expected error wording** is not in the spec; "required-field error", "incorrect-OTP error" and so on mean "a message with that meaning".